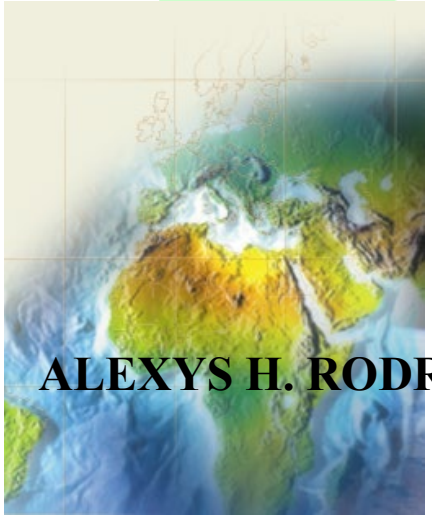
ESTRUCTURA DE DATOS VECTORIAL

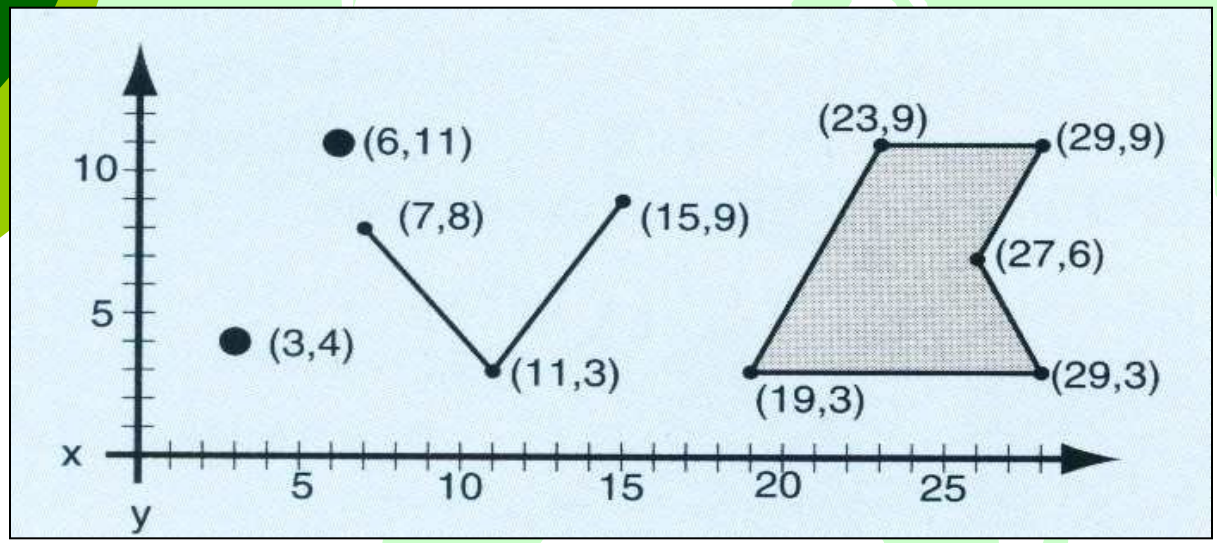


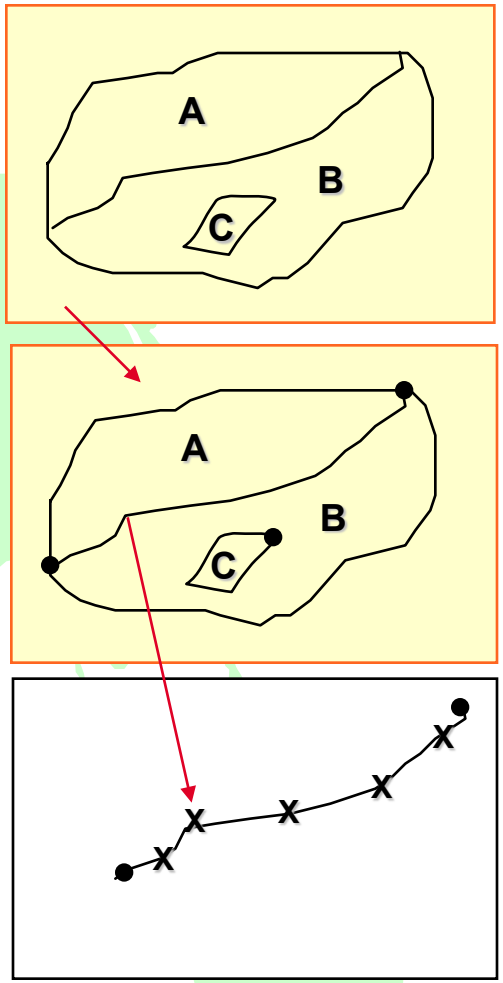
FUNDAMENTOS TEORICOS

ALEXYS H. RODRIGUEZ A. Ph.D.

MODELO VECTORIAL

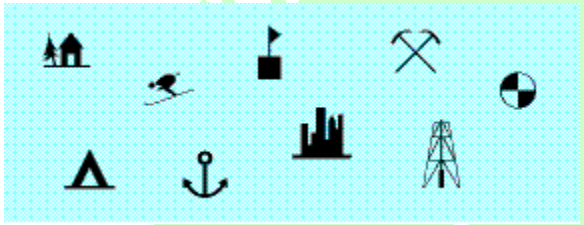
MODELO TOPOLÓGICO

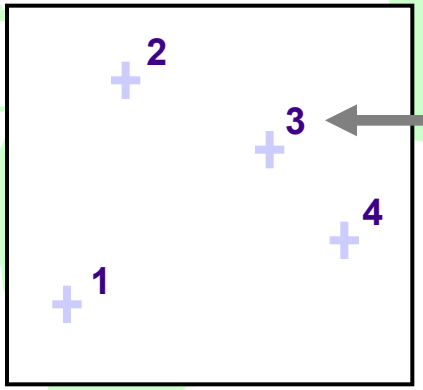




- INFORMACION ESTRUCTURADA (ARCOS Y NODOS)
- TABLAS DE RELACIONES TOPOLOGICAS (NODOS, AREAS, POLIGONOS)
- RELACIONES ESPACIALES DE: ADYACENCIA, CONECTIVIDAD Y PERTENENCIA
- ANALISIS INDEPENDIENTE DE LAS COORDENADAS

PUNTOS

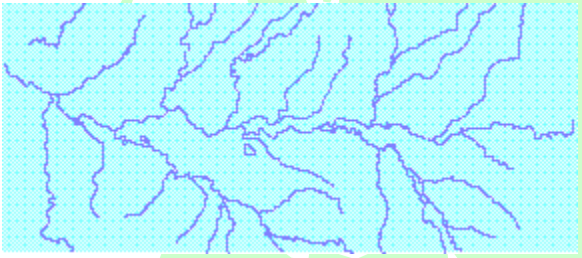


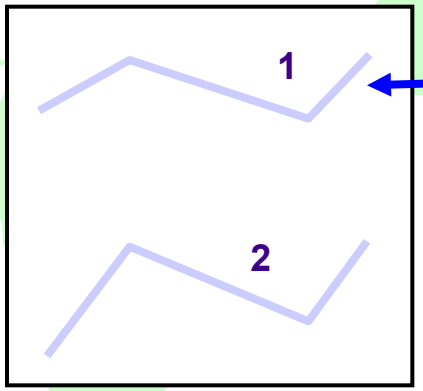


Número Punto	Coordenadas x,y
1	2,2
2	3,6
3	5,5
4	6,3

Representan entidades sin área ni longitud

LINEAS

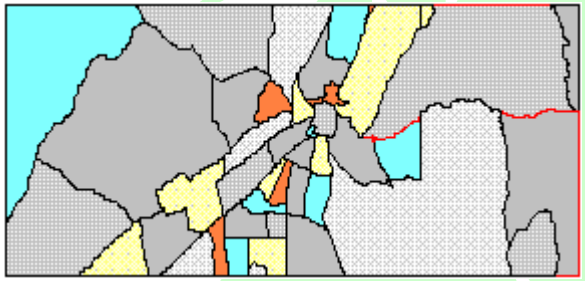


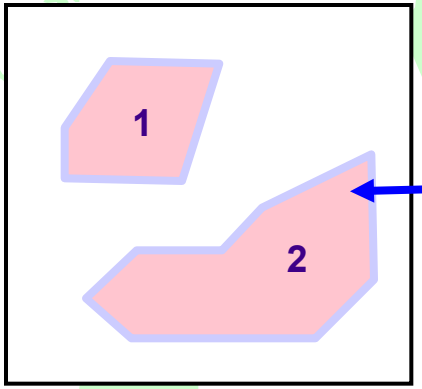


Número Línea	Coordenadas x,y
1	1,5 3,6 6,5 7 6
2	1,1 3,3 6,2 7,3

Representan entidades sin área pero con longitud

POLÍGONOS





Número Polígono	Coordenadas x,y
1	1,4 1,5 2,7 5,7 4,4 1,4
2	1,2 2,3 4,3 5,4 7,5 7,3 6,1 3,1 1,2

Las entidades representadas son áreas con características comunes

MODELO VECTORIAL

MODELO TOPOLÓGICO

ESTRUCTURA ARCO (Cadena de Segmentos Rectos) – NODO (Vértices donde cruzan dos o mas arcos)

TABLA 1: COORDENADAS DE LOS NODOS Y VÉRTICES DE CADA ARCO

Arco	Nodo Origen	Vértices Intermedios	Nodo Final
1	1, 6.5	5,8;6,8;6,9	5,6.5

TABLA 2: TOPOLOGÍA DE ARCOS

Arco	Nodo Origen	Nodo Final	Pol. Derecha	Pol. Izquierda
1	1	1	A	D

TABLA 3: TOPOLOGÍA DE POLÍGONOS

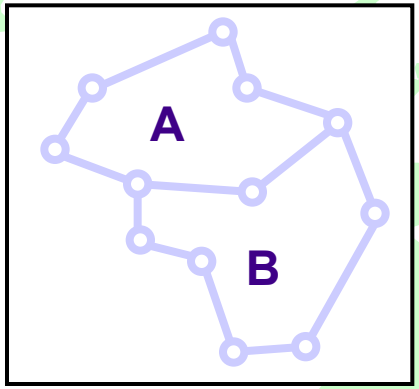
Polígono	Arcos
A	1,2,3,4,-15

TABLA 4: TOPOLOGÍA DE NODOS

Nodo	Arco
1	1,4,5

MODELO VECTORIAL

MODELO TOPOLÓGICO



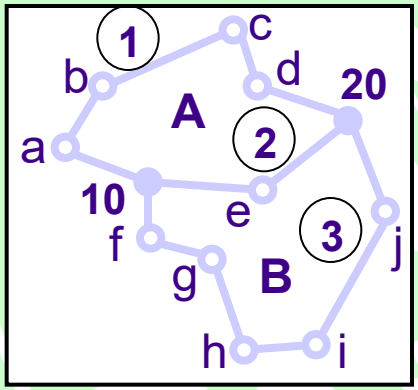


Tabla 1

Número arco	Nodo Inicial	Vértice	Nodo Final
1	20	d,c,b,a	10
2	10	e	20
3	10	f,g,h,i,j	20

Tabla 3

Polígono	Lista de arcos
A	1,2
B	2,3

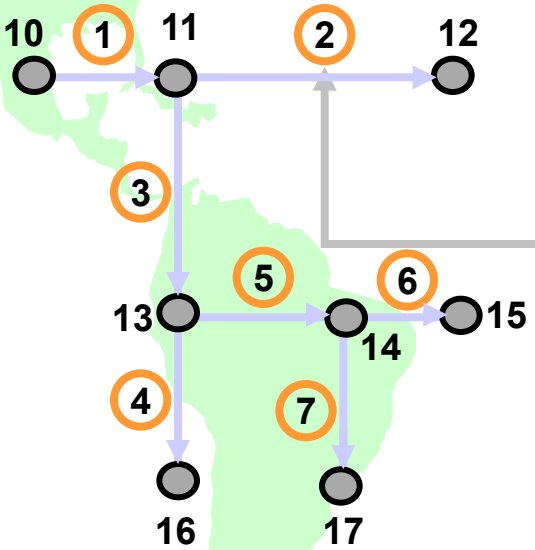
ESTRUCTURA ARCO – NODO

MODELO VECTORIAL

MODELO TOPOLÓGICO

CONECTIVIDAD, LONGITUD Y DIRECCIÓN

Topología Arco-nodo



Lista Arco-nodo

Arco	Del-Nodo	Al-Nodo
1	10	11
2	11	12
3	11	13
4	13	16
5	13	14
6	14	15
7	14	17

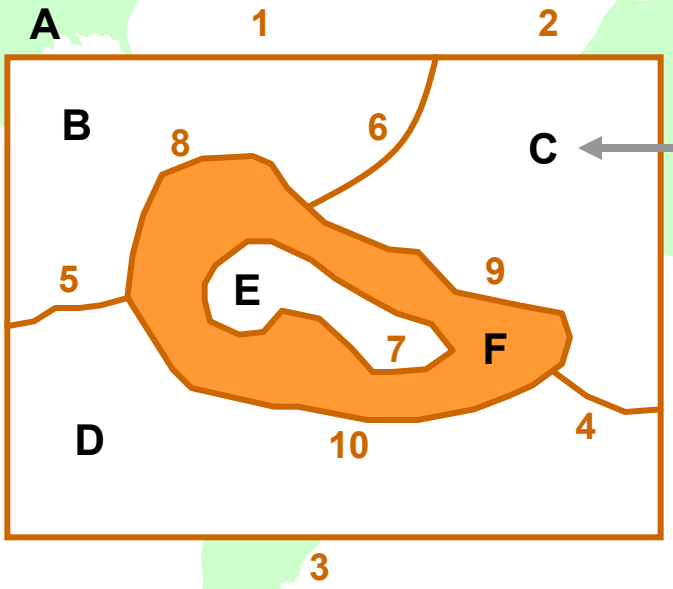
Tabla 2

ESTRUCTURA ARCO – NODO

MODELO VECTORIAL

MODELO TOPOLÓGICO

DEFINE LA CARACTERISTICA ESPACIAL DE UN ARCO DE PERTENECER A O SER COMPARTIDO POR DOS O MAS OBJETOS



Topología Polígono-Arco

Polígono	Lista de Arcos
B	1, 6, 8, 5
C	2, 4, 9, 6
D	3, 5, 10, 4
E	7
F	8, 9, 10, 0, 7

Tabla 3

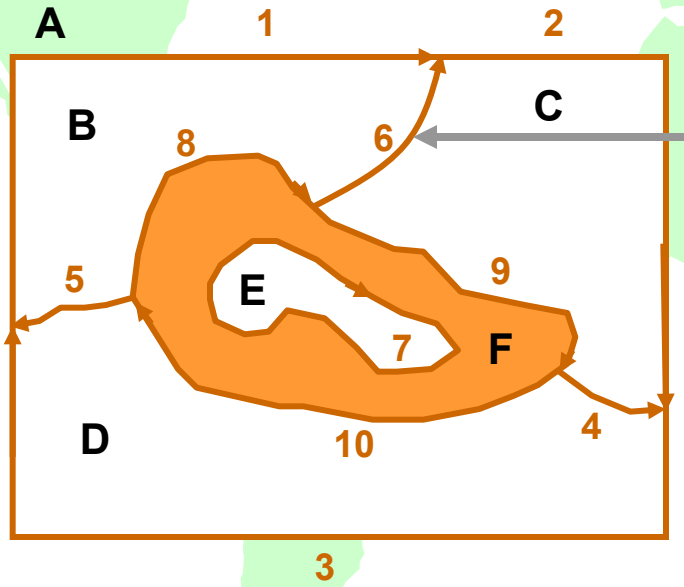
DEFINICIÓN DE ÁREA - PERTENENCIA

ESTRUCTURA ARCO – NODO

MODELO VECTORIAL

MODELO TOPOLÓGICO

ADYACENCIA



Topología Izquierda-Derecha

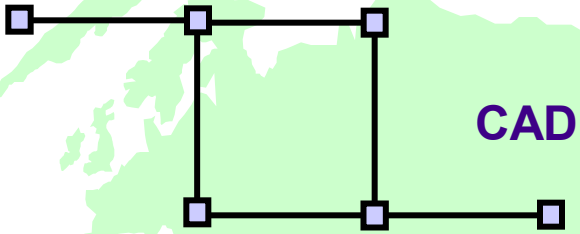
Arco	Polígono Izquierda	Polígono Derecha
1	A	B
2	A	C
3	A	D
4	C	D
5	D	B
6	B	C
7	F	E
8	B	F
9	C	F
10	D	F

Tabla 2

ESTRUCTURA ARCO – NODO

TRES MODELOS DE DATOS

- Dibujo

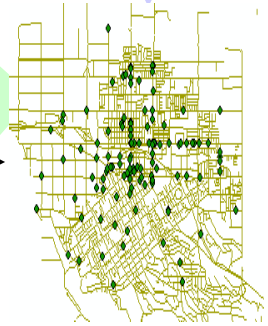


CAD

- GeoRelacional

Cobertura

LOCATION	OBJECTID	CASE_NUM	TYPE	REPORT_DA
145 S CHURCH ST	33	990302252	7	3/20/99
1711 N ORANGE ST	36	990100032	3	1/2/99
1702 N ORANGE ST	38	990302093	3	3/15/99
1144 OCCIDENTAL DR	53	990302239	6	3/20/99

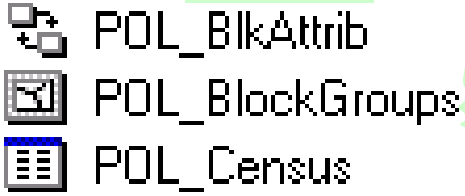


Tablas de atributos de elementos

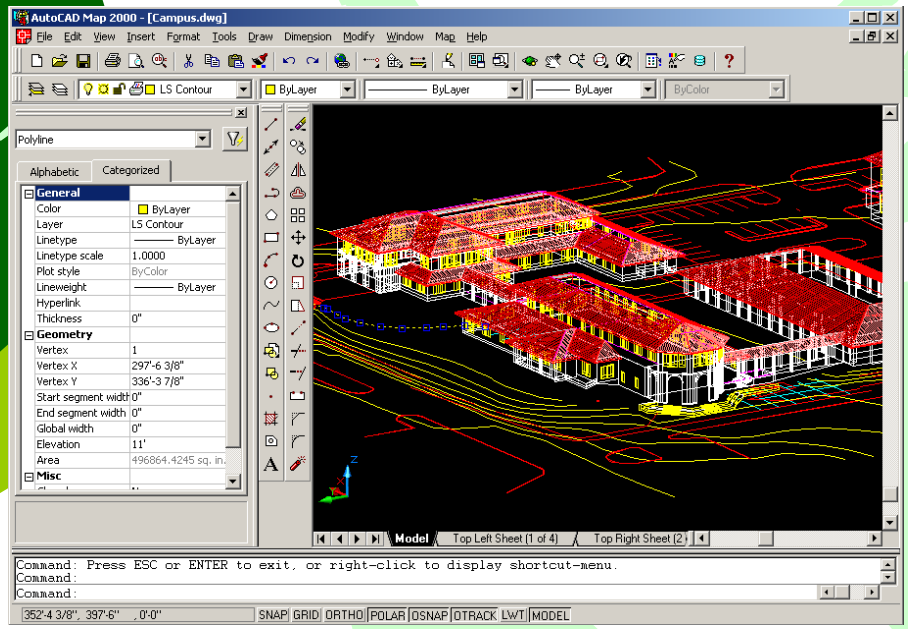
Archivos de cobertura

- Geodatabase

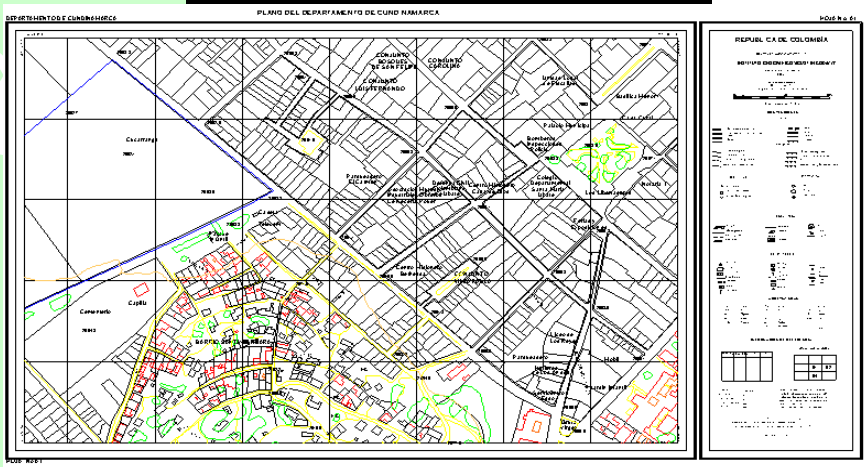
Objeto = dato + comportamiento



MODELO DE DATOS - CAD



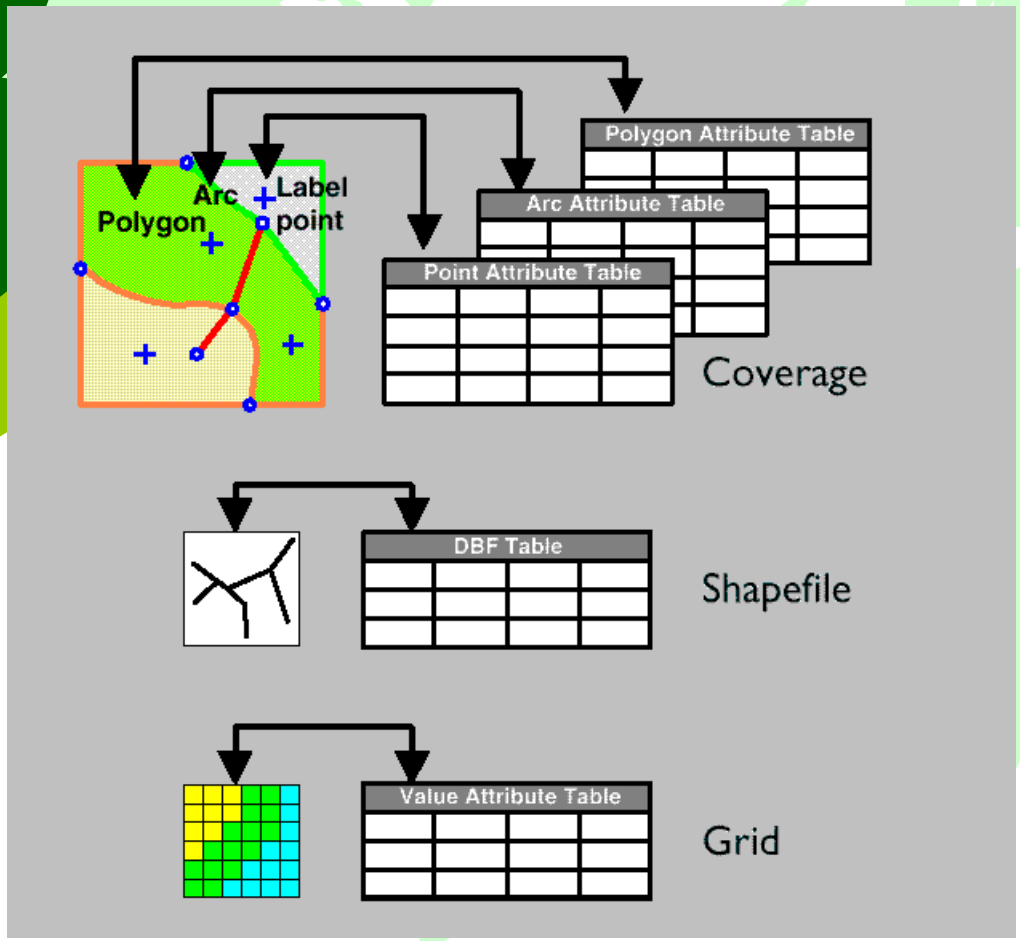
MAPA



PRIMERA GENERACIÓN: EN ESTA ERA (1960S Y 1970S), LOS MAPAS FUERON CREADOS CON SOFWARE CAD (COMPUTER-AIDED DESIGN) DE PROPÓSITO GENERAL. LAYERS Y ANOTACIONES FUERON LA REPRESNTACIÓN PRIMARIA DE ATRIBUTOS.

LOS ESCASOS DATOS ALFANUMÉRICOS SE ALMACENABAN COMO PROPIEDADES DE LAS ENTIDADES GEOGRÁFICAS, DENTRO DEL DIBUJO DIGITAL EN FORMATO BINARIO

MODELOS DE DATOS - GEORELACIONAL

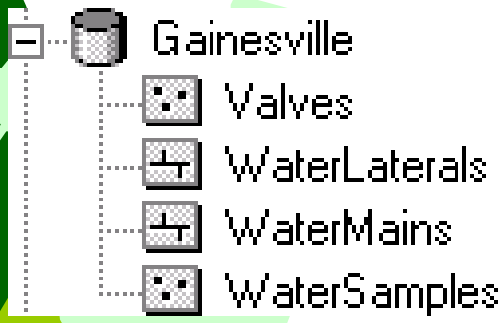


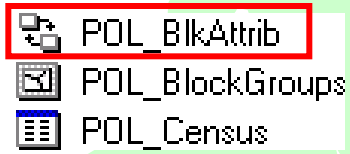
SEGUNDA GENERACIÓN - 1980s:

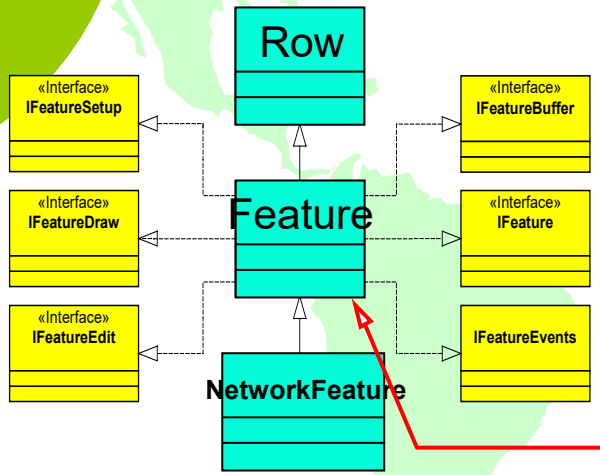
- DATOS ESPACIALES SON COMBINADOS CON DATOS DE ATRIBUTOS.
- ARCHIVOS BINARIOS INDEXADOS
- ALMACENAR RELACIONES TOPOLÓGICAS

VENTAJA: PERSONALIZAR LAS TABLAS DE ELEMENTOS GEOGRÁFICOS (AGREGAR CAMPOS Y ESTABLECER RELACIONES A TABLAS EN BASES DE DATOS EXTERNAS.

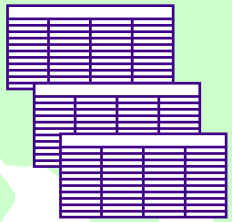
MODELOS DE DATOS - GEODATABASE

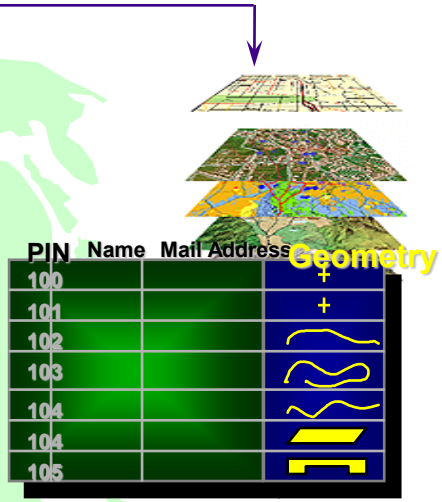


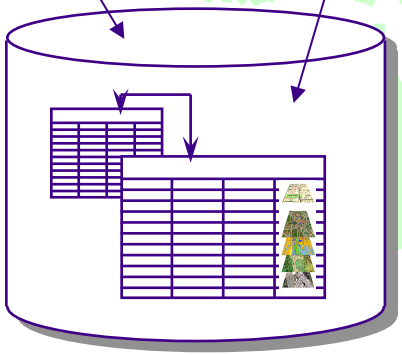




Objetos







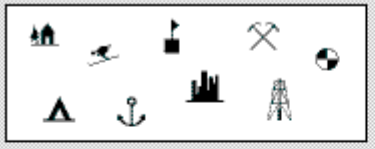
Objeto = dato + comportamiento

GEORELACIONAL - SHAPEFILE

.shx

.shp

.dbf



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2	114	mine
3	245	city
4	043	school
5	171	port

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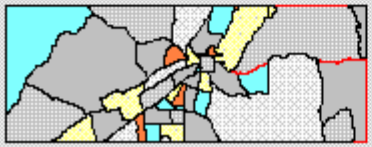
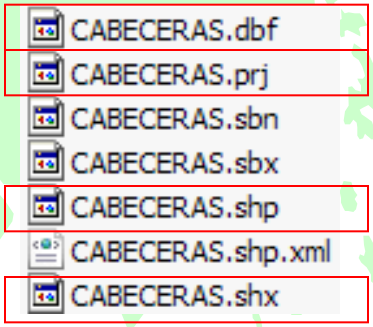
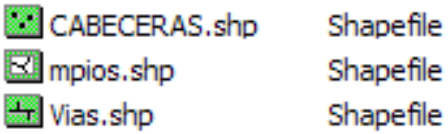
2	6	8	43	72	1
3	3	2	66	17	1
4	7	3	39	56	2
5	9	1	96	12	2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3	104	sedimentary
4	105	basalt
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